

PING

Koło Naukowe
Politechniki Gdańskiej



SHARK JACK CABLE

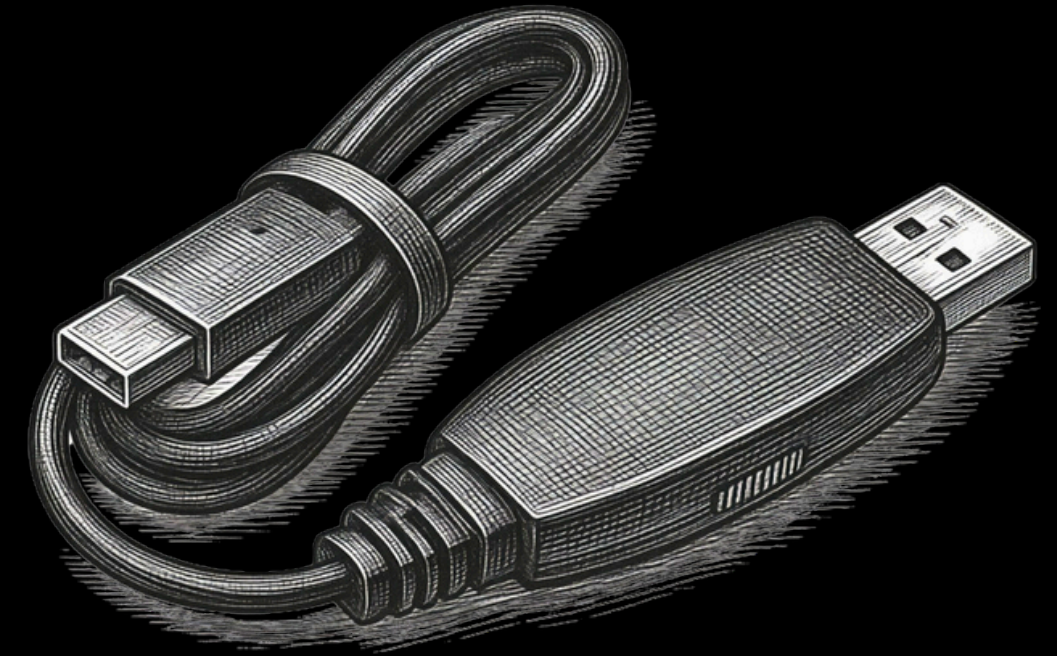
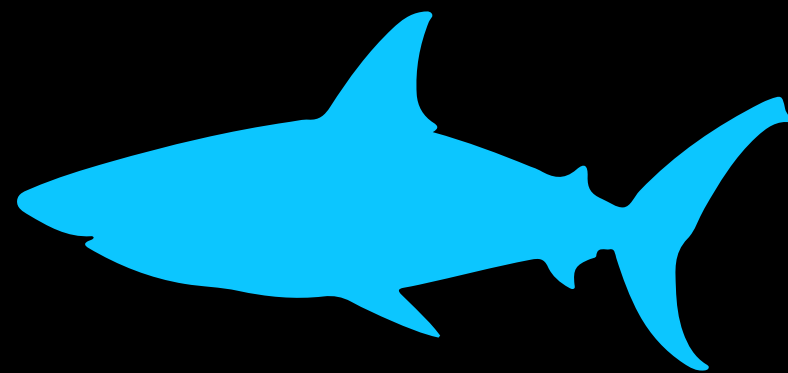
Pentesting device from Hak5



What is a Shark Jack?

Shark Jack in general is a **portable network attack and automation tool** for system administrators and pentesters.

It's build is small and includes additional LED lights to signalize different modes and actions.



It works as a **hotplug**, similarly to Rubber Ducky, scanning and attacking the network, but it can do many different operations like ex-filtration, data copying, reconnaissance and more. Thanks to LAN port it can quietly do anything without being verified, as almost all Ethernet networks are trusted without any doubt.



Shark Jack Cable vs Classic



There are two different variants of Shark Jack:

- **Shark Jack (classic one)** → It's original SJ with battery lasting from 10-15 minutes and feedback via RGB LED
- **Shark Jack cable** → Shark Jack cable is powered via USB-C, allowing it to run much longer as long as there is a source of power. It can be also connected to android terminal app for more elegant shell interaction.

This one I tested ->



How does it work?



A lot of technical details for this one....

The Shark Jack Cable operates as a `state-based Linux computer`.

In `Arming Mode ( USB-C connected )`, it simultaneously runs a DHCP Server, SSH Server, and Serial Server, (`set as NETMODE DHCP_SERVER`) allowing full management access.

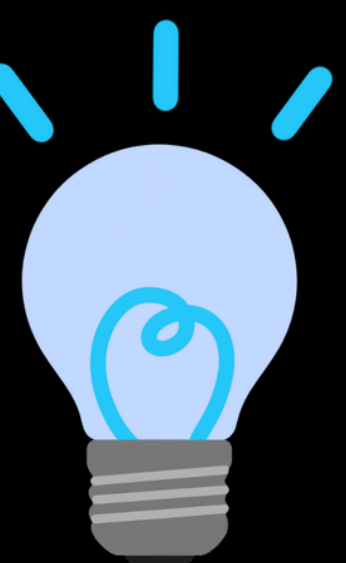
In `Attack Mode ( RJ45 connected to target )`, it automatically executes `/root/payload/payload.sh`, while acting as a DHCP Server to the target network.

When `NETMODE DHCP_CLIENT` is called inside a payload, it transitions to '`Guest`' mode, blending into the target network as a legitimate device with full internet/network access.



Few words about NETMODE

Shark Jack, beside it's modes, operates on two different, so called, `netmodes`.

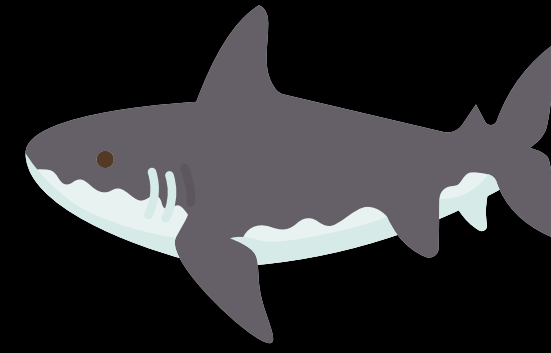
- 
- `NETMODE DHCP_SERVER` → It basically make shark jack acting as a server giving IP address to connected devices, it is also set as default mode during attack and boot.
 - `NETMODE DHCP_CLIENT` → It makes shark jack act as a guest inside the network, allowing for Internet access. Can be set ether when connected to the router for management reasons or during payload execution.

Both of those modes are set via `serial` or `ssh interface` while in arming mode.

IMPORTANT!: If SJ is connected in arming mode to the router and set as client, it can be access by any device within the same network using SSH protocol.



How to use it?



There are 2 main modes for SJ:

- **Attack mode** → When ready, once plugged in, it will run the payload and once completed the attack it will shine green
- **Arming mode** → This mode allows to edit payload files, catalogs, extensions and update the software / libraries.
 - In Shark Jack cable version you can use **serial USB terminal app** on android to connect it via USB-C and edit SJ files using linux like terminal or use **MobaXterm** app on PC (windows) to do the same.
 - You can also connect SJC via RJ45 to PC's port or router with Internet connection to modify files, update and more via SSH connection.

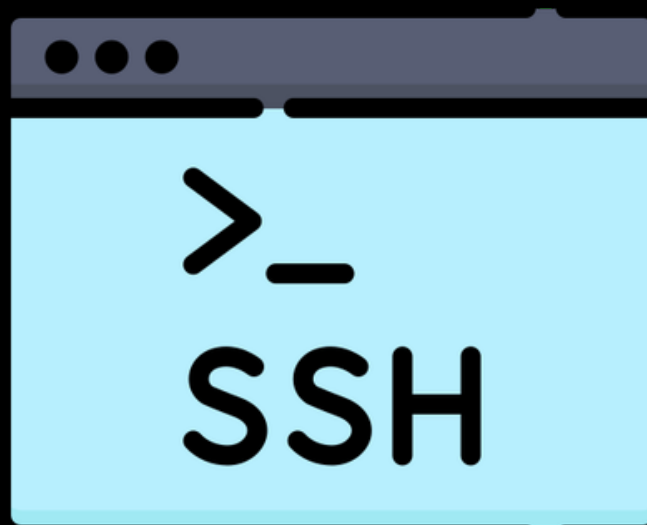
DISCLAIMER!: While connected via USB-C, Shark Jack won't have internet connection. If connected via RJ45 to router directly it will gain access to it and will be able to perform updates, curls and more, which can be viewed on serial terminal on PC or Android



How to connect to Shark Jack Cable using SSH protocol?

By default, SJC is supposed to be able act as `serial bridge controller` and network adapter once connected via USB-C, meaning that there should be a possibility to use SSH in this case. However there are multiple issues with that, especially on windows, which result in USB-C being only for powering and Serial Console access.

(`More details in report file`)



To connect via SSH you have to either use RJ45 end and plug it into your PC, or as mention before, via Router. In case of Router, one can use `ifconfig` command on serial console to get the IP and connect using `ssh root@<SHARK_IP>` or use SCP commands for copying



Introduction to the script

The script I prepared is a [Network Recon and Aggressive Scanning tool](#)

Upon starting the attack mode, Shark Jack runs the payload, sets [DHCP_CLIENT](#) to blend within the network and uses pre-installed tools like [nmap](#) and [tcpdump](#) to perform wide scanning of the target network, including traffic, data, IP and MAC addresses

Originally, there was supposed to be an ARP spoofing script as well, but due to infrastructure issues with SJC, I had to abandon the idea ([more details in report file](#))



payload.sh part 1

```
#!/bin/bash

# TITLE: Network Recon and Aggressive Scanning
# AUTHOR: Szymon Stefański ( KN PING GDAŃSK )
# DESCRIPTION: This script performs a network scan to discover alive hosts
# and runs aggressive scans on the first discovered host. It also captures
# some passive network intelligence
# VERSION: 2.0

LOOT_DIR_SCAN="/root/loot/scan_results"
LOOT_DIR_AUDIT="/root/loot/audit_results"
SCAN_TIME=$(date +%Y-%m-%d_%H-%M-%S)
mkdir -p $LOOT_DIR_SCAN
mkdir -p $LOOT_DIR_AUDIT

# >>>> SETUP <<<<

# Setting up client mode for stealth and Internet access

LED SETUP
SERIAL_WRITE "[*] Configuring network..." # SERIAL_WRITE is used to send messages to the serial console
NETMODE DHCP_CLIENT
sleep 10

# Gathering network information and Shark Jack's IP address

IP=""
TIMEOUT=0
while [ -z "$IP" ]; do
    IP=$(ifconfig eth0 | grep 'inet' | awk -F: '{print $2}' | awk '{print $1}')
    # we check interface eth0 for the assigned IP address
    # then using grep we select IPv4 line
    # awk is used to extract the IP address from the output
    sleep 2
    TIMEOUT=$((TIMEOUT + 2))
    if [ $TIMEOUT -ge 30 ]; then
        SERIAL_WRITE "[!] Failed to obtain IP address."
        LED FAIL
        exit 1
    fi
done
```

```
# We set the subnet based on the Shark Jack's IP, assuming a /24 network
# Save network info to network_info.txt

echo "===== " > $LOOT_DIR_SCAN/network_info.txt
echo "Shark Jack Network Report" >> $LOOT_DIR_SCAN/network_info.txt
echo "Scan Time: $SCAN_TIME" >> $LOOT_DIR_SCAN/network_info.txt
echo "===== " >> $LOOT_DIR_SCAN/network_info.txt
echo "SJC IP: $IP" >> $LOOT_DIR_SCAN/network_info.txt
echo "Gateway: $GATEWAY" >> $LOOT_DIR_SCAN/network_info.txt
echo "Subnet: $SUBNET" >> $LOOT_DIR_SCAN/network_info.txt
echo "===== " >> $LOOT_DIR_SCAN/network_info.txt

SERIAL_WRITE "[*] IP: $IP"
SERIAL_WRITE "[*] Gateway: $GATEWAY"
SERIAL_WRITE "[*] Subnet: $SUBNET"

# >>>> HOST DISCOVERY AND DEEP SCANNING <<<<

# Host discovery and deep scanning

LED ATTACK
SERIAL_WRITE "[*] Discovering alive hosts..."
nmap -sn $SUBNET -oG $LOOT_DIR_SCAN/alive_hosts.txt

# Count alive hosts ( we exclude the gateway and the Shark Jack itself using grep -v )

HOST_COUNT=$(grep "Up" $LOOT_DIR_SCAN/alive_hosts.txt | grep -v "$GATEWAY" | grep -v "$IP" | wc -l)
ALIVE_HOSTS=$(grep "Up" $LOOT_DIR_SCAN/alive_hosts.txt | grep -v "$GATEWAY" | grep -v "$IP" | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+')

if [ -z "$ALIVE_HOSTS" ]; then
    SERIAL_WRITE "[!] No targets found on the network."
    LED FAIL
    exit 1
fi

SERIAL_WRITE "[*] Deep port scanning all alive hosts..."
nmap -sV -O -T4 --top-ports 100 $SUBNET -oN $LOOT_DIR_SCAN/deep_scan.txt
sleep 5
nmap -sV -A $GATEWAY -oN $LOOT_DIR_SCAN/gateway_scan.txt
```



payload.sh part 2

```
# >>>> AGGRESSIVE SCANNING <<<<<

# We take the first alive host, better for testing reason
# For production / real use it is recommended to implement logic to select the most interesting target
# based on open ports and services or any other criteria.
# For example based on deep_scan.txt and gateway_scan we could select hosts with specific services
# and / or ports open.

TARGET_IP=$(echo "$ALIVE_HOSTS" | head -n 1)
SERIAL_WRITE "[*] Aggressive scan on $TARGET_IP..."

# Aggressive scan with OS detection, version detection, script scanning and traceroute
# Limiting to top 20 ports, for speed and storage reasons on Shark Jack

nmap -sV -O -A --osscan-guess --top-ports 20 $TARGET_IP -oN $LOOT_DIR_AUDIT/detailed_audit_$TARGET_IP.txt

# >>>> TRAFFIC CAPTURE <<<<<

# Passive traffic capture for network intelligence

SERIAL_WRITE "[*] Capturing broadcast traffic (60s)..."

# Capture MDNS (device names and services)

tcpdump -i eth0 -vv -t udp port 5353 -c 100 > $LOOT_DIR_SCAN/mdns_devices.txt 2>&1 &

# Capture ARP traffic (who is talking to whom)

tcpdump -i eth0 -n arp -c 100 > $LOOT_DIR_SCAN/arp_traffic.txt 2>&1 &

# Capture any unencrypted HTTP traffic

tcpdump -i eth0 -A -l -c 100 'tcp port 80' > $LOOT_DIR_SCAN/http_traffic.txt 2>&1 &

sleep 60
```

```
# >>>> CLEANUP AND FINAL REPORT <<<<<

# Clean up and final report
SERIAL_WRITE "[*] Generating final report..."
LED CLEANUP

killall tcpdump 2>/dev/null
killall nmap 2>/dev/null

# Generate summary report
echo "===== " > $LOOT_DIR_SCAN/SUMMARY.txt
echo "SCAN SUMMARY" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "===== " >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "Date: $SCAN_TIME" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "SJC IP: $IP" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "Gateway: $GATEWAY" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "Subnet: $SUBNET" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "Hosts Found: $HOST_COUNT" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "Primary Target: $TARGET_IP" >> $LOOT_DIR_SCAN/SUMMARY.txt
echo "===== " >> $LOOT_DIR_SCAN/SUMMARY.txt

SERIAL_WRITE "[*] Scan complete. Results saved to $LOOT_DIR_SCAN and $LOOT_DIR_AUDIT"
SERIAL_WRITE "[*] Files generated:"
SERIAL_WRITE "    - network_info.txt"
SERIAL_WRITE "    - alive_hosts.txt"
SERIAL_WRITE "    - deep_scan.txt"
SERIAL_WRITE "    - gateway_scan.txt"
SERIAL_WRITE "    - detailed_audit_*.txt"
SERIAL_WRITE "    - mdns_devices.txt"
SERIAL_WRITE "    - arp_traffic.txt"
SERIAL_WRITE "    - http_traffic.txt"
SERIAL_WRITE "    - SUMMARY.txt"

LED FINISH
```



Screens from testing

```
PS C:\Users\... \DeepScan> scp -r -o "UserKnownHostsFile=/dev/null" -o "StrictHostKeyCheck
ing=no" root@...:./
Warning: Permanently added '...' (ED25519) to the list of known hosts.
root@...s password:
detailed_audit_...txt 100% 2474 172.6KB/s 00:00
http_traffic.txt      100% 650 79.4KB/s 00:00
SUMMARY.txt          100% 290 35.4KB/s 00:00
network_info.txt     100% 263 23.4KB/s 00:00
arp_traffic.txt       100% 1858 151.2KB/s 00:00
deep_scan.txt         100% 36KB 315.2KB/s 00:00
alive_hosts.txt       100% 1432 139.8KB/s 00:00
mdns_devices.txt      100% 21KB 1.4MB/s 00:00
... \DeepScan> |
```

SCAN SUMMARY

Date: 2021-08-23_20-02-02

SJC IP: [REDACTED]

fe80

Gateway: [REDACTED]

Subnet: [REDACTED]

Hosts Found: 18

Primary Target: [REDACTED]

Shark Jack Network Report

Scan Time: 2021-08-23_20-02-02

SJC IP: [REDACTED]

fe80

Gateway: [REDACTED]

Subnet: [REDACTED]



Screens from testing

Terminal

Terminal

BusyBox v1.28.4 () built-in shell (ash)

Shark Jack

by Hak5

Type HELP for usage

root@shark:/#

root@shark:/# [51.035714] rt3050-esw 10110000.esw: link changed 0x01

NETMODE DHCP_CLIENT

NETMODE DHCP_CLIENT

root@shark:/#

root@shark:/# ping -c3 example.com

ping -c3 example.com

PING example.com (172.66.147.243): 56 data bytes

64 bytes from 172.66.147.243: seq=0 ttl=58 time=22.406 ms

64 bytes from 172.66.147.243: seq=1 ttl=58 time=19.951 ms

64 bytes from 172.66.147.243: seq=2 ttl=58 time=19.654 ms

example.com ping statistics ---

3 packets transmitted, 3 packets received, 0% packet loss

round-trip min/avg/max = 19.654/20.670/22.406 ms

root@shark:/#

root@shark:/#

M1

M2

M3

M4

M5

M6

M7

>

root@shark:~#

root@shark:~# [*] Starting Payload

[*] Configuring network...

[*] IP: 192.168.0.241 fe80

[*] Gateway:

[*] Subnet:

[*] Discovering alive hosts...

[*] Deep port scanning all alive hosts...

[*] Aggressive scan on

[*] Capturing broadcast traffic (60s)...

[1964.153724] device eth0 entered promiscuous mode

[*] Generating final report...

[2023.856900] device eth0 left promiscuous mode

[*] Scan complete. Results saved to /root/loot/scan_results and /root/loot/audit_results

[*] Files generated:

- network_info.txt

- alive_hosts.txt

- deep_scan.txt

- gateway_scan.txt

- detailed_audit_*.txt

- mdns_devices.txt

- arp_traffic.txt

- http_traffic.txt

- SUMMARY.txt

M1

M2

M3

M4

M5

M6

M7

>

Ochrona sieci

Czas	Zdarzenie	Dział...	Obiekt...	Obiekt...	P..	Reguła/ ...	Ścieżka apl...	Apli...	Hash	Uży...	Syg...	Naz...	Ust...	
10.05.202...	Wykryto atak...	Zablo...	192.168...	192.168...	T...	Win32/Bo...								



References



- [Github Repo for Shark Jack payloads](#)
- [Hak5 Shark Jack documentation](#)
- [Useful commands for copy](#)
- [A little more about NETMODE](#)
- [Serial Console showcase \(Android\)](#)
- [MobaXterm](#)

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Thank you kindly for your time!

Prepared by Szymon Stefański

